

INTERVAL ESTIMATES

INTERVAL ESTIMATION

- Suppose that $X_1, \dots, X_n$ is a sample from a normal population having unknown mean μ and known variance σ^2
- It has been shown that $\bar{X} = \sum_{i=1}^n X_i / n$ is the maximum likelihood estimator for μ
- The sample mean $\bar{X}$ will not be exactly equal to μ
- Rather than a point estimate, it is sometimes more valuable to be able to specify an interval for which we have a certain degree of confidence that μ lies within

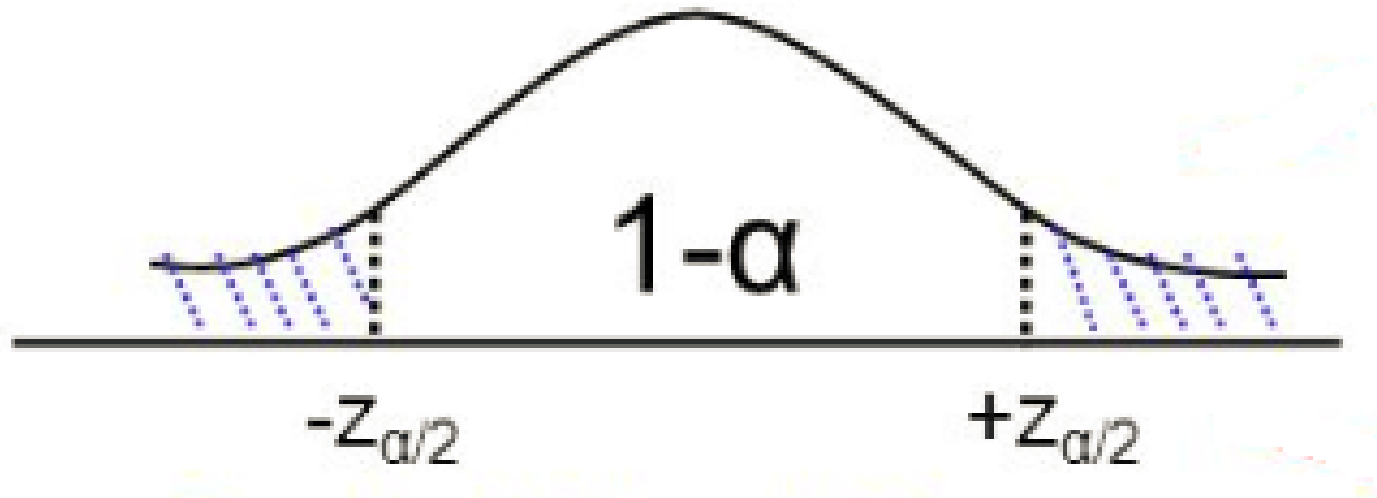
100(1- α)% confidence interval

- Suppose that a sample $X_1, X_2, \dots, X_n$ is sample from a normal distribution with unknown mean μ and unknown variance σ^2
- Further suppose that $L_1(X_1, \dots, X_n)$ and $L_2(X_1, \dots, X_n)$ are two statistics such that $L_1 < L_2$ with probability 1
- The interval (L_1, L_2) is called a $[100(1-\alpha)]\%$ *confidence interval* for L_1 and L_2 such that

$$P(L_1 < \theta < L_2) = 1 - \alpha.$$

- Limits L_1 and L_2 are called, respectively, the *lower* and *upper confidence limits* for θ , and $1 - \alpha$ is called the *confidence coefficient*
- The value of $1 - \alpha$ is generally taken as 0.90, 0.95 and 0.99 ($\alpha = .1, 0.05, 0.01$)

95% confidence interval



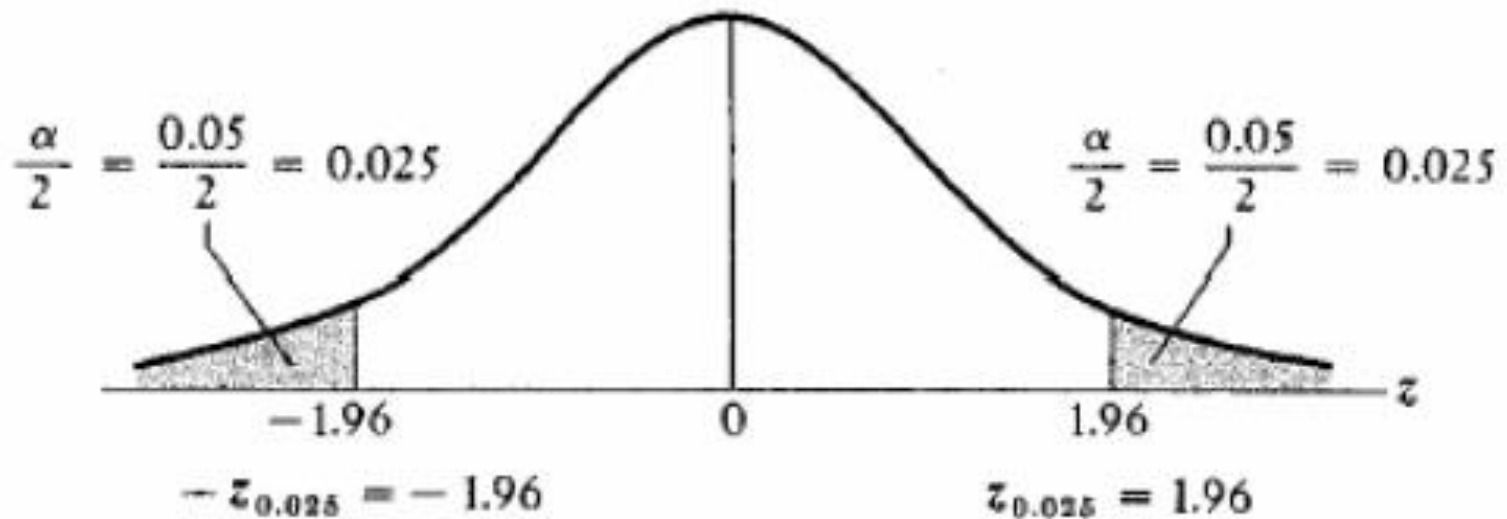
$$\alpha=0.05$$

$$1-\alpha=.95$$

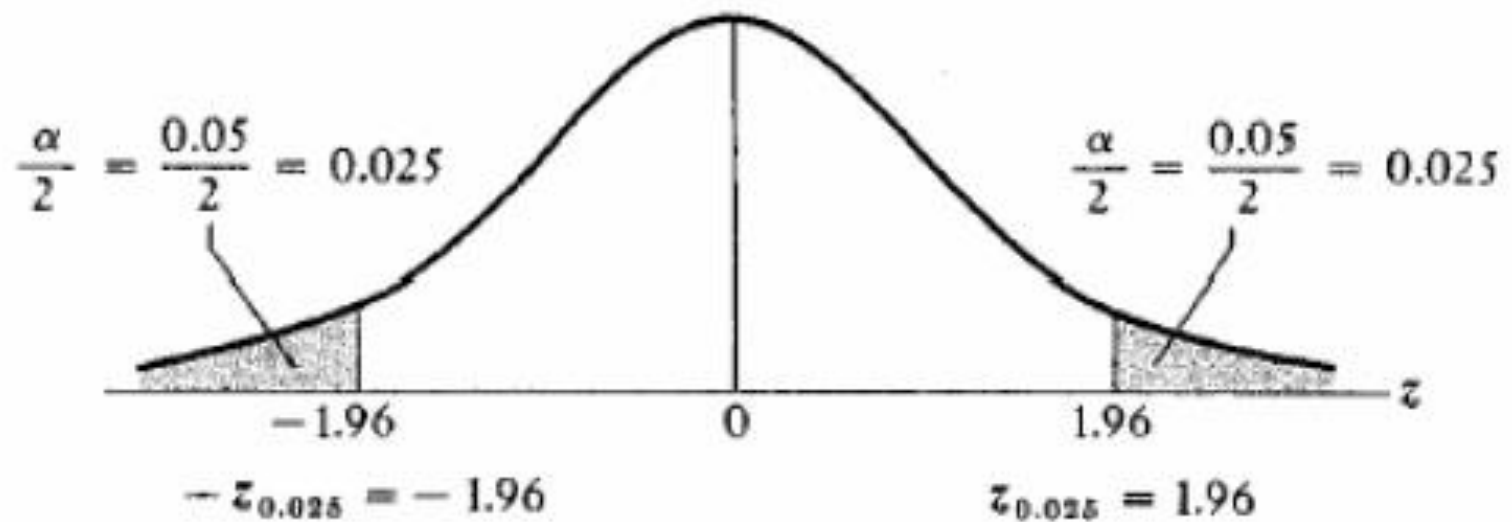
$$\alpha/2=0.025$$

$$Z_{\alpha/2} = 1.96$$

$100(1-\alpha)\%$ confidence interval



- α is the rejection area



$$P(-u_{\alpha/2} < U < u_{\alpha/2}) = 1 - \alpha.$$

$$P(-1.96 < U < 1.96) = \int_{-1.96}^{1.96} f_U(u) du = 0.95,$$

$$P \left\{ -1.96 < \sqrt{n} \frac{(\bar{X} - \mu)}{\sigma} < 1.96 \right\} = .95$$

- Multiplying with $\sigma/\sqrt{n}$

$$P \left\{ -1.96 \frac{\sigma}{\sqrt{n}} < \bar{X} - \mu < 1.96 \frac{\sigma}{\sqrt{n}} \right\} = .95$$

- Multiplying with -1 throughout

$$P \left\{ -1.96 \frac{\sigma}{\sqrt{n}} < \mu - \bar{X} < 1.96 \frac{\sigma}{\sqrt{n}} \right\} = .95$$

$$P \left\{ \bar{X} - 1.96 \frac{\sigma}{\sqrt{n}} < \mu < \bar{X} + 1.96 \frac{\sigma}{\sqrt{n}} \right\} = .95$$

95 percent confidence

- “with 95 percent confidence” we assert that the true mean lies within $1.96\sigma/\sqrt{n}$ of the observed sample mean
- The interval

$$\left(\bar{x} - 1.96 \frac{\sigma}{\sqrt{n}}, \bar{x} + 1.96 \frac{\sigma}{\sqrt{n}} \right)$$

is called a *95 percent confidence interval estimate* of μ .

Problem

- Suppose that when a signal having value x is transmitted from location A the value received at location B is normally distributed with mean μ and variance 4. That is, if x is sent, then the value received is $x + N$ where N , representing noise, is normal with mean 0 and variance 4. To reduce error, suppose the same value is sent 9 times. If the successive values received are 5, 8.5, 12, 15, 7, 9, 7.5, 6.5, 10.5, construct a 95 percent confidence interval for x .

Two-sided confidence interval

- *95 percent confidence interval estimate of x*

$$\left(\bar{x} - 1.96 \frac{\sigma}{\sqrt{n}}, \bar{x} + 1.96 \frac{\sigma}{\sqrt{n}} \right)$$

$$\bar{x} = \frac{81}{9} = 9$$

$$\left(9 - 1.96 \frac{\sigma}{3}, 9 + 1.96 \frac{\sigma}{3} \right) = (7.69, 10.31)$$

- we are “95 percent confident” that the true message value lies between 7.69 and 10.31

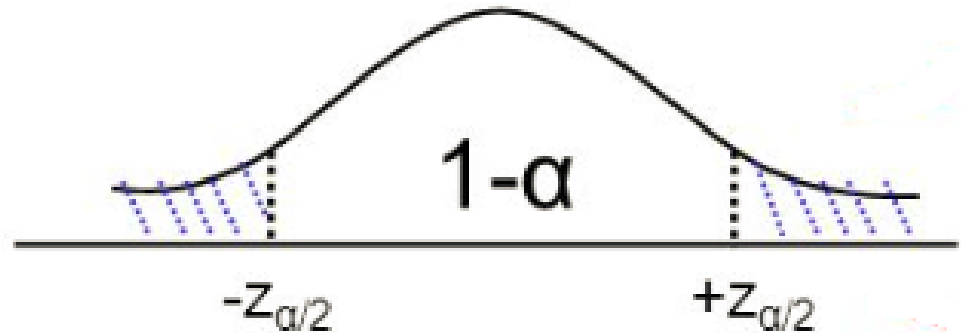
95 percent *one-sided upper confidence interval* for μ

- Sometimes we are interested in determining a value so that we can assert with, say, 95 percent confidence, that μ is at least as large as that value
- To determine such a value, note that if Z is a standard normal random variable then

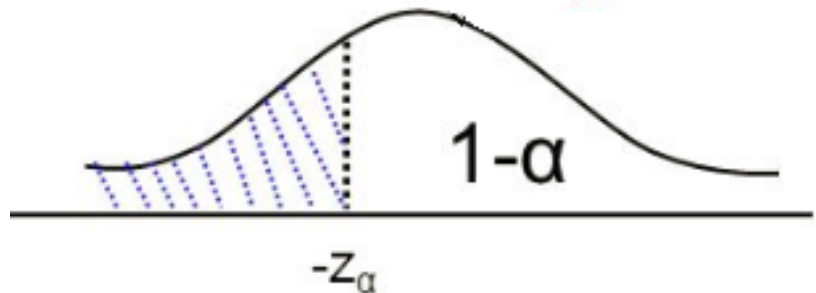
$$P\{Z < 1.645\} = .95$$

95 percent *one-sided upper and lower confidence* *confidence*

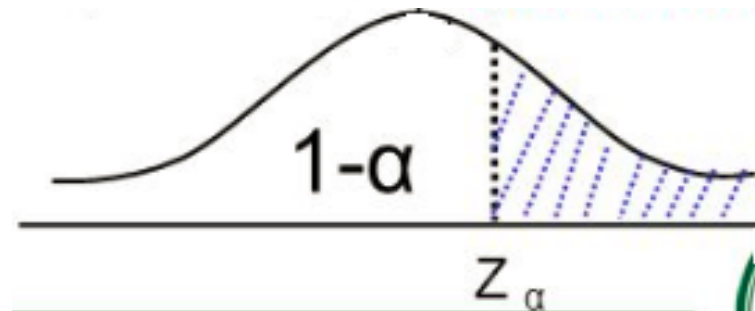
- Two sided interval



- One sided upper interval



- One sided lower interval



95 percent *one-sided upper confidence interval* for μ

$$P\{Z < 1.645\} = .95$$

$$P\left\{\sqrt{n}\frac{(\bar{X} - \mu)}{\sigma} < 1.645\right\} = .95$$

$$P\left\{\bar{X} - 1.645\frac{\sigma}{\sqrt{n}} < \mu\right\} = .95$$

- 95 percent *one-sided upper confidence interval* for μ is

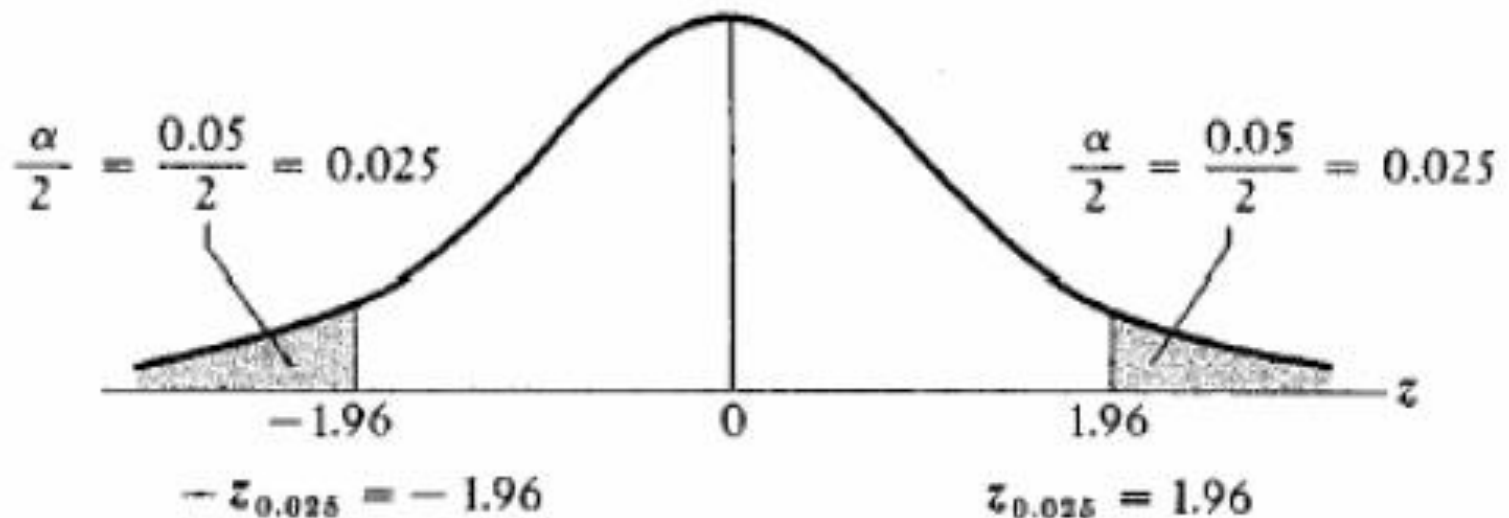
$$\left(\bar{x} - 1.645\frac{\sigma}{\sqrt{n}}, \infty\right)$$

One-sided lower confidence interval

$$\left(\bar{x} - 1.96 \frac{\sigma}{\sqrt{n}}, \bar{x} + 1.96 \frac{\sigma}{\sqrt{n}} \right)$$

- The observed value of the sample mean is $\bar{x}$, then the 95 percent one-sided lower confidence interval for μ is

$$\left(-\infty, \bar{x} + 1.645 \frac{\sigma}{\sqrt{n}} \right)$$



- Interval: $\left(\bar{x} - 1.96 \frac{\sigma}{\sqrt{n}}, \bar{x} + 1.96 \frac{\sigma}{\sqrt{n}} \right)$

$$\left(-\infty, \bar{x} + 1.645 \frac{\sigma}{\sqrt{n}} \right)$$

$$\left(\bar{x} - 1.645 \frac{\sigma}{\sqrt{n}}, \infty \right)$$

- Lower confidence upper confidence

Determine the upper and lower 95 percent confidence interval estimates of μ in previous Example

$$1.645 \frac{\sigma}{\sqrt{n}} = \frac{3.29}{3} = 1.097$$

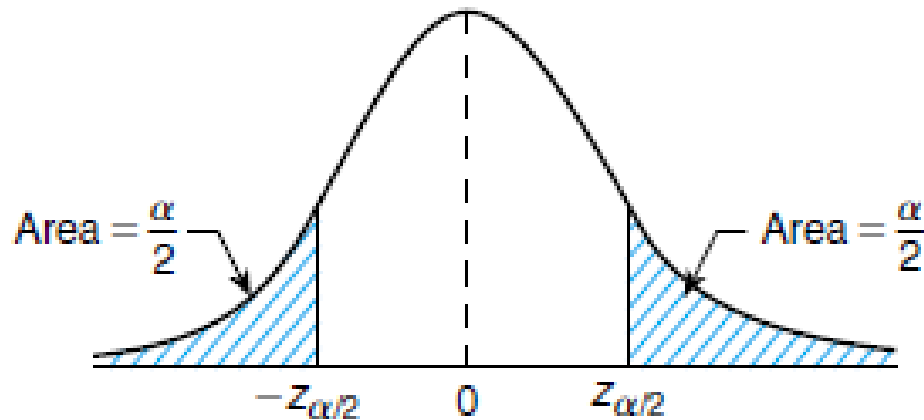
the 95 percent upper confidence interval is

$$(9 - 1.097, \infty) = (7.903, \infty)$$

and the 95 percent lower confidence interval is

$$(-\infty, 9 + 1.097) = (-\infty, 10.097)$$

Confidence intervals of any specified level of confidence



$$P\{-z_{\alpha/2} < Z < z_{\alpha/2}\} = 1 - \alpha$$

$$P\left\{\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} < \mu < \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}\right\} = 1 - \alpha$$

- $100(1 - \alpha)$ percent two-sided confidence interval for μ

$$\left(\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \quad \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right)$$

- one-sided confidence intervals of any desired level of confidence

- Upper confidence $\left(\bar{x} - z_{\alpha} \frac{\sigma}{\sqrt{n}}, \infty \right)$

- Lower confidence $\left(-\infty, \bar{x} + z_{\alpha} \frac{\sigma}{\sqrt{n}} \right)$

- Obtain a 99 percent confidence interval estimate of μ , along with 99 percent one-sided upper and lower intervals
- Since $z_{.005} = 2.58$

$$2.58 \frac{\sigma}{\sqrt{n}} = \frac{5.16}{3} = 1.72$$

That is, the 99 percent confidence interval estimate is (7.28, 10.72).

since $z_{.01} = 2.33$, a 99 percent upper confidence interval is

$$(9 - 2.33(2/3), \infty) = (7.447, \infty)$$

Similarly, a 99 percent lower confidence interval is

$$(-\infty, 9 + 2.33(2/3)) = (-\infty, 10.553)$$

To choose the sample size n so that the interval is of a certain size

- Suppose that we want to compute an interval of length .1 that we can assert, with 99 percent confidence for the previous problem containing μ . How large need n be?
- Interval for μ from a sample of size n is

$$\left(\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \quad \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right)$$

- Its length is : $2 * z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$

- 99 percent confidence interval for μ from a sample of size n is

$$\left(\bar{x} - 2.58 \frac{\sigma}{\sqrt{n}}, \quad \bar{x} + 2.58 \frac{\sigma}{\sqrt{n}} \right)$$

- its length is $5.16 \frac{\sigma}{\sqrt{n}}$
- To make the length of the interval equal to .1,

$$5.16 \frac{\sigma}{\sqrt{n}} = .1 \qquad n = (51.6\sigma)^2$$

- From past experience it is known that the weights of salmon grown at a commercial hatchery are normal with a mean that varies from season to season but with a standard deviation that remains fixed at 0.3 pounds. If we want to be 95 percent certain that our estimate of the present season's mean weight of a salmon is correct to within ± 0.1 pounds, how large a sample is needed?

- A 95 percent confidence interval estimate for the unknown mean μ , based on a sample of size n , is

$$\mu \in \left(\bar{x} - 1.96 \frac{\sigma}{\sqrt{n}}, \bar{x} + 1.96 \frac{\sigma}{\sqrt{n}} \right)$$

- We can be 95 percent certain that $\bar{x}$ is within ± 0.1 of μ provided that

$$2 \times 1.96 \times \frac{\sigma}{\sqrt{n}} = 0.2$$

$$2 \times 1.96 \times \frac{0.3}{\sqrt{n}} = 0.2$$

$$\frac{1.176}{\sqrt{n}} = 0.2$$

$$n = 34.57$$

- That is, a sample size of 35 or larger will suffice.

t distribution

- Sample sizes are sometimes small, and often we do not know the standard deviation of the population
- When either of these problems occur, statisticians rely on the distribution of the **t statistic**(also known as the **t score**), whose values are given by: $\frac{\bar{X} - \mu}{s/\sqrt{n}}$

$$\text{Where } s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

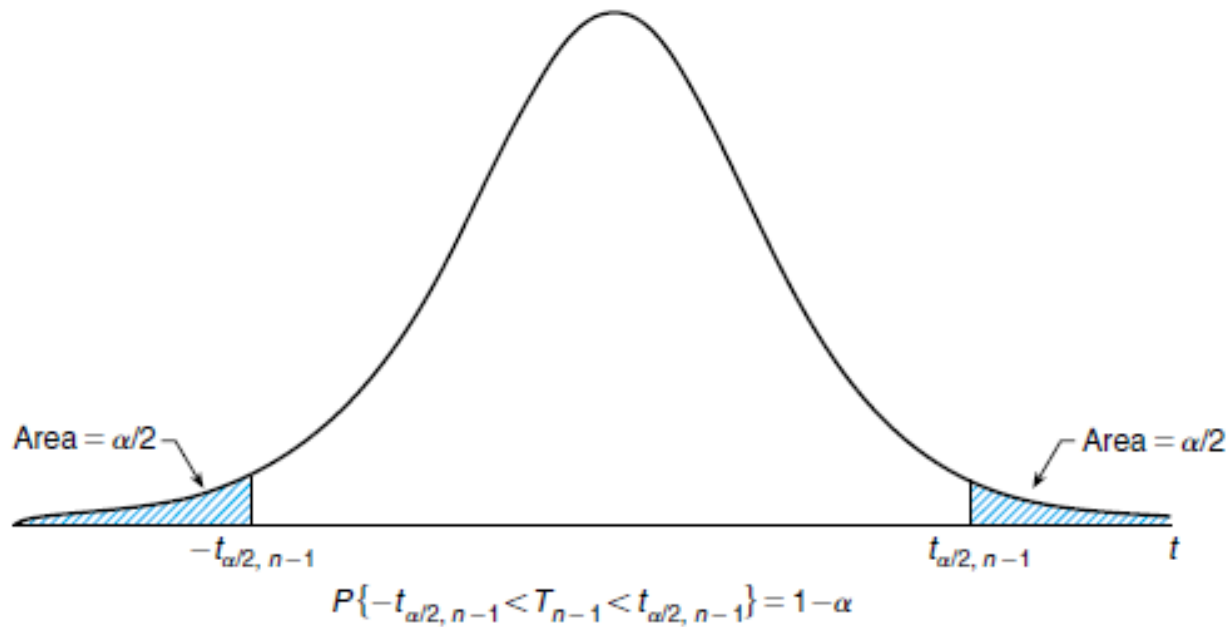
When should we use the t distribution?

- You must **use** the **t-distribution** table when working with problems when the population standard deviation (σ) is not known and the sample size is small ($n < 30$).
- If σ is not known, then **using t-distribution** is correct. If σ is known, then **using the normal distribution** is correct.

Confidence Interval for a Normal Mean When the Variance Is Unknown

- Suppose now that $X_1, \dots, X_n$ is a sample from a normal distribution with unknown mean μ and unknown variance σ^2 , and that we wish to construct a $100(1-\alpha)$ percent confidence interval for μ
- From t distribution: $t = \frac{\bar{X} - \mu}{s/\sqrt{n}}$ is a t -random variable with $n - 1$ degrees of freedom

$100(1 - \alpha)$ percent confidence



$$P \left\{ -t_{\alpha/2, n-1} < \sqrt{n} \frac{(\bar{X} - \mu)}{S} < t_{\alpha/2, n-1} \right\} = 1 - \alpha$$

$100(1 - \alpha)$ percent confidence

$$P \left\{ -t_{\alpha/2, n-1} < \sqrt{n} \frac{(\bar{X} - \mu)}{S} < t_{\alpha/2, n-1} \right\} = 1 - \alpha$$

$$P \left\{ \bar{X} - t_{\alpha/2, n-1} \frac{S}{\sqrt{n}} < \mu < \bar{X} + t_{\alpha/2, n-1} \frac{S}{\sqrt{n}} \right\} = 1 - \alpha$$

- Thus, if it is observed that $\bar{X} = \bar{x}$ and $S = s$, then we can say that “with $100(1 - \alpha)$ percent confidence”

$$\mu \in \left(\bar{x} - t_{\alpha/2, n-1} \frac{s}{\sqrt{n}}, \bar{x} + t_{\alpha/2, n-1} \frac{s}{\sqrt{n}} \right)$$

Problem

- The value x is transmitted from location A then the value received at location B is normal with mean μ and variance σ^2 but with σ^2 being unknown. If 9 successive values are, as in 5, 8.5, 12, 15, 7, 9, 7.5, 6.5, 10.5 compute a 95 percent confidence interval for μ .

TABLE A3 *Values of $t_{\alpha,n}$*

n	$\alpha = .10$	$\alpha = .05$	$\alpha = .025$	$\alpha = .01$	$\alpha = .005$
1	3.078	6.314	12.706	31.821	63.657
2	1.886	2.920	4.303	6.965	9.925
3	1.638	2.353	3.182	4.541	5.841
4	1.533	2.132	2.776	3.474	4.604
5	1.476	2.015	2.571	3.365	4.032
6	1.440	1.943	2.447	3.143	3.707
7	1.415	1.895	2.365	2.998	3.499
8	1.397	1.860	2.306	2.896	3.355
9	1.383	1.833	2.262	2.821	3.250
10	1.372	1.812	2.228	2.764	3.169
11	1.363	1.796	2.201	2.718	3.106
12	1.356	1.782	2.179	2.681	3.055
13	1.350	1.771	2.160	2.650	3.012
14	1.345	1.761	2.145	2.624	2.977
15	1.341	1.753	2.131	2.602	2.947
16	1.337	1.746	2.120	2.583	2.921
17	1.333	1.740	2.110	2.567	2.898
18	1.330	1.734	2.101	2.552	2.878
19	1.328	1.729	2.093	2.539	2.861
20	1.325	1.725	2.086	2.528	2.845
21	1.323	1.721	2.080	2.518	2.831
22	1.321	1.717	2.074	2.508	2.819
23	1.319	1.714	2.069	2.500	2.807
24	1.318	1.711	2.064	2.492	2.797
25	1.316	1.708	2.060	2.485	2.787

Solution

$$\bar{x} = 9$$

$$s^2 = \frac{\sum x_i^2 - 9(\bar{x})^2}{8} = 9.5$$

Hence, as $t_{.025,8} = 2.306$, a 95 percent confidence interval for μ is

$$\left[9 - 2.306 \frac{(3.082)}{3}, 9 + 2.306 \frac{(3.082)}{3} \right] = (6.63, 11.37)$$

One- sided confidence intervals

- $100(1 - \alpha)$ *upper* confidence interval would be

$$\mu \in \left(\bar{x} - \frac{s}{\sqrt{n}} t_{\alpha, n-1}, \infty \right)$$

- $100(1 - \alpha)$ *lower* confidence interval would be

$$\mu \in \left(-\infty, \bar{x} + \frac{s}{\sqrt{n}} t_{\alpha, n-1} \right)$$

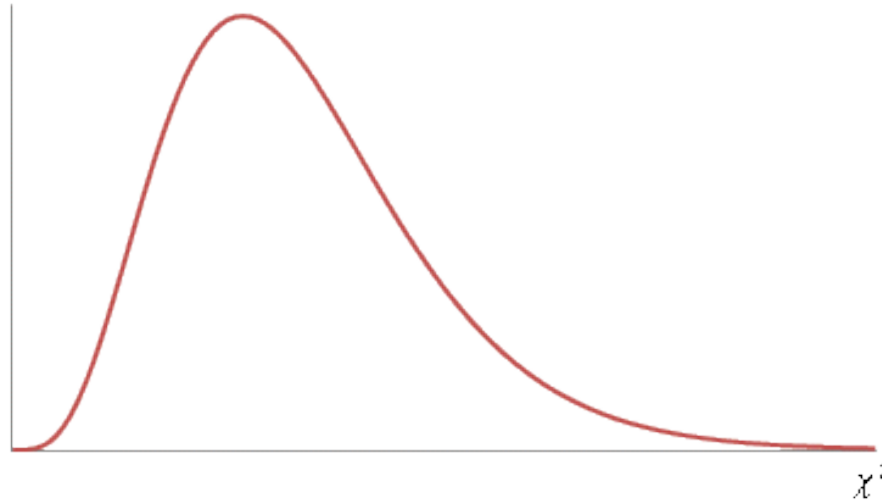
Chi Square distribution

- The Chi Square distribution is the distribution of the sum of squared *standard normal deviates*
- If a simple random sample size n is obtained from a normally distributed population with mean μ and standard deviation σ , then

$$\chi^2 = \frac{(n-1) s^2}{\sigma^2}$$

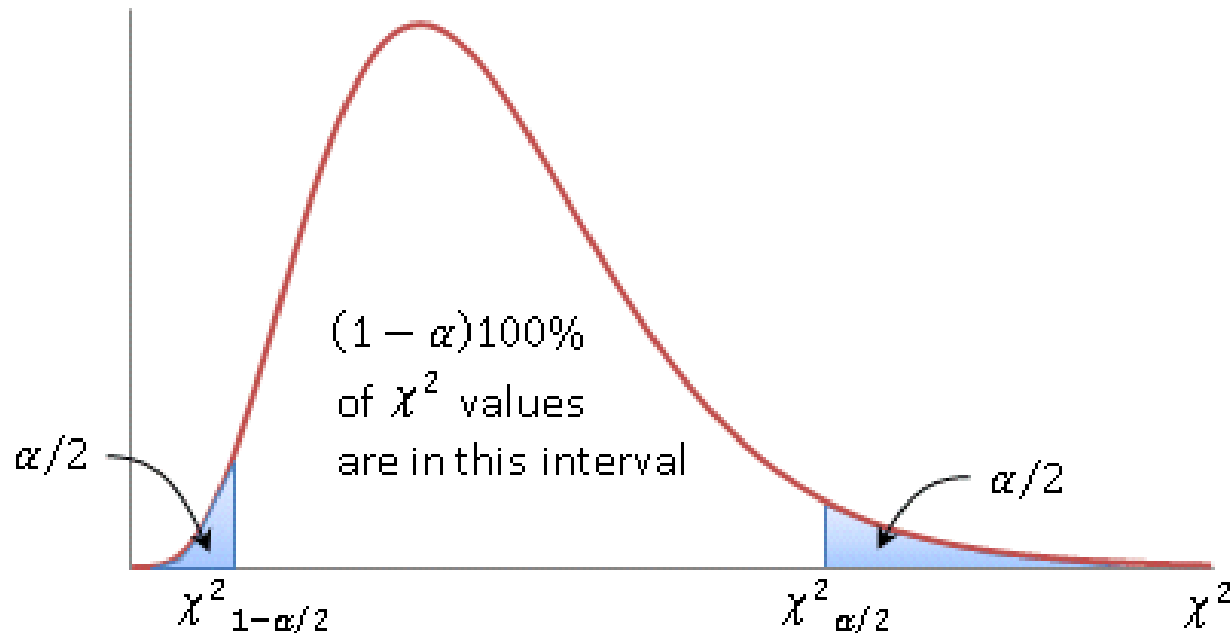
has a **chi-square distribution** with $n-1$ degrees of freedom.

Chi Square distribution



- **Properties of the χ^2 distribution**
- It is *not* symmetric.
- The shape depends on the degrees of freedom.
- As the number of degrees of freedom increases, the distribution becomes more symmetric.
- $\chi^2 \geq 0$

$(1-\alpha)$ of the χ^2



$$\chi^2_{1-\alpha/2} < \frac{(n-1)s^2}{\sigma^2} < \chi^2_{\alpha/2}$$

Confidence Intervals for the Variance of a Normal Distribution

- If $X_1, \dots, X_n$ is a sample from a normal distribution having unknown parameters μ and σ^2 , then we can construct a confidence interval for σ^2 by using the fact that

$$\chi^2 = \frac{(n-1) s^2}{\sigma^2}$$

$$\text{where } s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

100(1 - α) percent confidence interval for σ^2

$$P \left\{ \chi_{1-\alpha/2, n-1}^2 \leq (n-1) \frac{S^2}{\sigma^2} \leq \chi_{\alpha/2, n-1}^2 \right\} = 1 - \alpha$$

$$P \left\{ \frac{(n-1)S^2}{\chi_{\alpha/2, n-1}^2} \leq \sigma^2 \leq \frac{(n-1)S^2}{\chi_{1-\alpha/2, n-1}^2} \right\} = 1 - \alpha$$

Hence when $S^2 = s^2$, a 100(1 - α) percent confidence interval for σ^2 is

$$\left\{ \frac{(n-1)s^2}{\chi_{\alpha/2, n-1}^2}, \frac{(n-1)s^2}{\chi_{1-\alpha/2, n-1}^2} \right\}$$

Problem

- A standardized procedure is expected to produce washers with very small deviation in their thicknesses. Suppose that 10 such washers were chosen and measured. If the thicknesses of these washers were, in inches,

.123 .133

.124 .125

.126 .128

.120 .124

.130 .126

what is a 90 percent confidence interval for the standard deviation of the thickness of a washer produced by this procedure?

TABLE A2 Values of $\chi^2_{\alpha,n}$

n	$\alpha = .995$	$\alpha = .99$	$\alpha = .975$	$\alpha = .95$	$\alpha = .05$	$\alpha = .025$	$\alpha = .01$	$\alpha = .005$
1	.0000393	.000157	.000982	.00393	3.841	5.024	6.635	7.879
2	.0100	.0201	.0506	.103	5.991	7.378	9.210	10.597
3	.0717	.115	.216	.352	7.815	9.348	11.345	12.838
4	.207	.297	.484	.711	9.488	11.143	13.277	14.860
5	.412	.554	.831	1.145	11.070	12.832	13.086	16.750
6	.676	.872	1.237	1.635	12.592	14.449	16.812	18.548
7	.989	1.239	1.690	2.167	14.067	16.013	18.475	20.278
8	1.344	1.646	2.180	2.733	15.507	17.535	20.090	21.955
9	1.735	2.088	2.700	3.325	16.919	19.023	21.666	23.589
10	2.156	2.558	3.247	3.940	18.307	20.483	23.209	25.188
11	2.603	3.053	3.816	4.575	19.675	21.920	24.725	26.757
12	3.074	3.571	4.404	5.226	21.026	23.337	26.217	28.300
13	3.565	4.107	5.009	5.892	22.362	24.736	27.688	29.819
14	4.075	4.660	5.629	6.571	23.685	26.119	29.141	31.319
15	4.601	5.229	6.262	7.261	24.996	27.488	30.578	32.801
16	5.142	5.812	6.908	7.962	26.296	28.845	32.000	34.267
17	5.697	6.408	7.564	8.672	27.587	30.191	33.409	35.718
18	6.265	7.015	8.231	9.390	28.869	31.526	34.805	37.156
19	6.844	7.633	8.907	10.117	30.144	32.852	36.191	38.582
20	7.434	8.260	9.591	10.851	31.410	34.170	37.566	39.997
21	8.034	8.897	10.283	11.591	32.671	35.479	38.932	41.401
22	8.643	9.542	10.982	12.338	33.924	36.781	40.289	42.796
23	9.260	10.196	11.689	13.091	35.172	38.076	41.638	44.181
24	9.886	10.856	12.401	13.844	36.415	39.364	42.980	45.558
25	10.520	11.524	13.120	14.611	37.652	40.646	44.314	46.928

Solution

$$S^2 = 1.366 \times 10^{-5}$$

Because $\chi^2_{.05,9} = 16.917$ and $\chi^2_{.95,9} = 3.334$, and because

$$\frac{9 \times 1.366 \times 10^{-5}}{16.917} = 7.267 \times 10^{-6},$$

$$\frac{9 \times 1.366 \times 10^{-5}}{3.334} = 36.875 \times 10^{-6}$$

Solution

it follows that, with confidence .90,

$$\sigma^2 \in (7.267 \times 10^{-6}, \quad 36.875 \times 10^{-6})$$

Taking square roots yields that, with confidence .90,

$$\sigma \in (2.696 \times 10^{-3}, \quad 6.072 \times 10^{-3})$$

100(1 - α) percent confidence interval

$$X_1, \dots, X_n \sim \mathcal{N}(\mu, \sigma^2)$$

$$\bar{X} = \sum_{i=1}^n X_i / n, \quad S = \sqrt{\sum_{i=1}^n (X_i - \bar{X})^2 / (n-1)}$$

Assumption	Parameter	Confidence Interval	Lower Interval	Upper Interval
σ^2 known	μ	$\bar{X} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$	$\left(-\infty, \bar{X} + z_{\alpha} \frac{\sigma}{\sqrt{n}}\right)$	$\left(\bar{X} - z_{\alpha} \frac{\sigma}{\sqrt{n}}, \infty\right)$
σ^2 unknown	μ	$\bar{X} \pm t_{\alpha/2, n-1} \frac{S}{\sqrt{n}}$	$\left(-\infty, \bar{X} + t_{\alpha, n-1} \frac{S}{\sqrt{n}}\right)$	$\left(\bar{X} - t_{\alpha, n-1} \frac{S}{\sqrt{n}}, \infty\right)$
μ unknown	σ^2	$\left(\frac{(n-1)S^2}{\chi_{\alpha/2, n-1}^2}, \frac{(n-1)S^2}{\chi_{1-\alpha/2, n-1}^2}\right)$	$\left(0, \frac{(n-1)S^2}{\chi_{1-\alpha, n-1}^2}\right)$	$\left(\frac{(n-1)S^2}{\chi_{\alpha, n-1}^2}, \infty\right)$

Estimating the difference in means of two normal populations

- Let $X_1, X_2, \dots, X_n$ be a sample of size n from a normal population having mean μ_1 and variance σ_1^2 and let $Y_1, \dots, Y_m$ be a sample of size m from a different normal population having mean μ_2 and variance σ_2^2 and suppose that the two samples are independent of each other
- We are interested in estimating $\mu_1 - \mu_2$

Estimating $\mu_1 - \mu_2$

- Since $\bar{X} = \sum_{i=1}^n X_i/n$ and $\bar{Y} = \sum_{i=1}^m Y_i/m$ are the maximum likelihood estimators of μ_1 and μ_2 then, $\bar{X} - \bar{Y}$ will be the maximum likelihood estimator of $\mu_1 - \mu_2$
- To obtain a confidence interval estimator, we need the distribution of $\bar{X} - \bar{Y}$

Estimating $\mu_1 - \mu_2$

- Because $\bar{X} \sim \mathcal{N}(\mu_1, \sigma_1^2/n)$
 $\bar{Y} \sim \mathcal{N}(\mu_2, \sigma_2^2/m)$
- Hence by additive property
$$\bar{X} - \bar{Y} \sim \mathcal{N}\left(\mu_1 - \mu_2, \frac{\sigma_1^2}{n} + \frac{\sigma_2^2}{m}\right)$$
- Assuming σ_1^2 and σ_2^2 are known, we have that

$$\frac{\bar{X} - \bar{Y} - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n} + \frac{\sigma_2^2}{m}}} \sim \mathcal{N}(0, 1)$$

Estimating $\mu_1 - \mu_2$

- So

$$P \left\{ -z_{\alpha/2} < \frac{\bar{X} - \bar{Y} - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n} + \frac{\sigma_2^2}{m}}} < z_{\alpha/2} \right\} = 1 - \alpha$$

- Equivalently

$$P \left\{ \bar{X} - \bar{Y} - z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n} + \frac{\sigma_2^2}{m}} < \mu_1 - \mu_2 < \bar{X} - \bar{Y} + z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n} + \frac{\sigma_2^2}{m}} \right\} = 1 - \alpha$$

Estimating $\mu_1 - \mu_2$

If X and Y are observed to equal x and y , respectively, then a $100(1-\alpha)$ two-sided confidence interval estimate for $\mu_1 - \mu_2$ is

$$\mu_1 - \mu_2 \in \left(\bar{x} - \bar{y} - z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n} + \frac{\sigma_2^2}{m}}, \bar{x} - \bar{y} + z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n} + \frac{\sigma_2^2}{m}} \right)$$

One-sided confidence intervals for $\mu_1 - \mu_2$ are obtained in a similar fashion

- Two different types of electrical cable insulation have recently been tested to determine the voltage level at which failures tend to occur. When specimens were subjected to an increasing voltage stress in a laboratory experiment, failures for the two types of cable insulation occurred at the following voltages:

<i>Type A</i>	<i>Type B</i>
36 54	52 60
44 52	64 44
41 37	38 48
53 51	68 4
38 44	66 70
36 35	52 62
34 44	

- Suppose that it is known that the amount of voltage that cables having type A insulation can withstand is normally distributed with unknown mean μ_A and known variance $\sigma_A^2 = 40$, whereas the corresponding distribution for type B insulation is normal with unknown mean μ_B and known variance $\sigma_B^2 = 100$. Determine a 95 percent confidence interval for $\mu_A - \mu_B$.

The 95% confidence interval for the mean is (219.6056,26.4897)

Estimating $\mu_1 - \mu_2$ with unknown variance

- Let us suppose now that we again desire an interval estimator of $\mu_1 - \mu_2$ but that the population variances σ_1^2 and σ_2^2 are unknown
- Replace σ_1^2 and σ_2^2 in equation by the sample variances S_1^2 and S_2^2

$$\frac{\bar{X} - \bar{Y} - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n} + \frac{\sigma_2^2}{m}}} \sim \mathcal{N}(0, 1)$$

$$S_1^2 = \sum_{i=1}^n \frac{(X_i - \bar{X})^2}{n-1}$$
$$S_2^2 = \sum_{i=1}^m \frac{(Y_i - \bar{Y})^2}{m-1}$$

Estimating $\mu_1 - \mu_2$ with unknown variance

- Hence the interval estimate will be

$$\frac{\bar{X} - \bar{Y} - (\mu_1 - \mu_2)}{\sqrt{S_1^2/n + S_2^2/m}}$$

Estimating $\mu_1 - \mu_2$ with unknown variance

- When the standard deviation of either population is unknown and the sample sizes (n and m) are large, the standard deviation of the sampling distribution can be estimated by the **standard error**, using the equation below

$$\sqrt{\frac{S_1^2}{n} + \frac{S_2^2}{m}}$$

Estimating $\mu_1 - \mu_2$ with unknown variance

- It is only in the special case when $\sigma_1^2 = \sigma_2^2$ that we will be able to obtain an interval estimator
- let us suppose that the population variances, though unknown, are equal and let σ^2 denote their common value

$$\frac{\bar{X} - \bar{Y} - (\mu_1 - \mu_2)}{\sqrt{\sigma^2/n + \sigma^2/m}} = \frac{\bar{X} - \bar{Y} - (\mu_1 - \mu_2)}{\sigma\sqrt{1/n + 1/m}}$$

- Since σ^2 is unknown this result cannot be immediately applied to obtain a confidence interval; σ^2 must first be estimated.
- To do so, note that both sample variances are estimators of σ^2 ; moreover, since S_1^2 has $n-1$ degrees of freedom and S_2^2 has $m-1$, the appropriate estimator is to take a weighted average of the two sample variances, with the weights proportional to these degrees of freedom

the estimator of σ^2 is the *pooled estimator*

$$S_p^2 = \frac{(n-1)S_1^2 + (m-1)S_2^2}{n+m-2}$$

Estimating $\mu_1 - \mu_2$ with unknown variance

- When the population standard deviations are unknown, but assumed to be equal; and the samples sizes n and m are small (under 30) and $\bar{X}_1, S_1^2, \bar{X}_2, S_2^2$ are independent random variables, the **standard error** for small population is given by

$$S_p^2 = \frac{(n-1)S_1^2 + (m-1)S_2^2}{n+m-2}$$

Estimating $\mu_1 - \mu_2$ with unknown variance

- Using t -distribution with $n + m - 2$ degrees of freedom, the interval is given by

$$P \left\{ -t_{\alpha/2, n+m-2} \leq \frac{\bar{X} - \bar{Y} - (\mu_1 - \mu_2)}{S_p \sqrt{1/n + 1/m}} \leq t_{\alpha/2, n+m-2} \right\} = 1 - \alpha$$

Therefore, when the data result in the values $\bar{X} = \bar{x}, \bar{Y} = \bar{y}, S_p = s_p$, we obtain the following $100(1 - \alpha)$ percent confidence interval for $\mu_1 - \mu_2$:

$$\left(\bar{x} - \bar{y} - t_{\alpha/2, n+m-2} s_p \sqrt{1/n + 1/m}, \quad \bar{x} - \bar{y} + t_{\alpha/2, n+m-2} s_p \sqrt{1/n + 1/m} \right)$$

Assumption

Confidence Interval

σ_1, σ_2 known

$$\bar{X} - \bar{Y} \pm z_{\alpha/2} \sqrt{\sigma_1^2/n + \sigma_2^2/m}$$

σ_1, σ_2 unknown but equal

$$\bar{X} - \bar{Y} \pm t_{\alpha/2, n+m-2} \sqrt{\left(\frac{1}{n} + \frac{1}{m}\right) \frac{(n-1)S_1^2 + (m-1)S_2^2}{n+m-2}}$$

Assumption

Lower Confidence Interval

σ_1, σ_2 known

$$(-\infty, \bar{X} - \bar{Y} + z_{\alpha} \sqrt{\sigma_1^2/n + \sigma_2^2/m})$$

σ_1, σ_2 unknown but equal

$$\left(-\infty, \bar{X} - \bar{Y} + t_{\alpha, n+m-2} \sqrt{\left(\frac{1}{n} + \frac{1}{m}\right) \frac{(n-1)S_1^2 + (m-1)S_2^2}{n+m-2}} \right)$$